PATHFIT 1

UNDERSTANDING MOVEMENTS

IDEAL POSTURE, MOVEMENT TERMS, DIRECTIONAL TERMS, PLANES OF MOTION, JOINTS

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POSTURE

"the **relative position of the body** and/or the **arrangement of its body parts** at any moment."



EVOLUTION

respond to the **challenges**
in the environment

PRESSURE

spine + muscles and joints

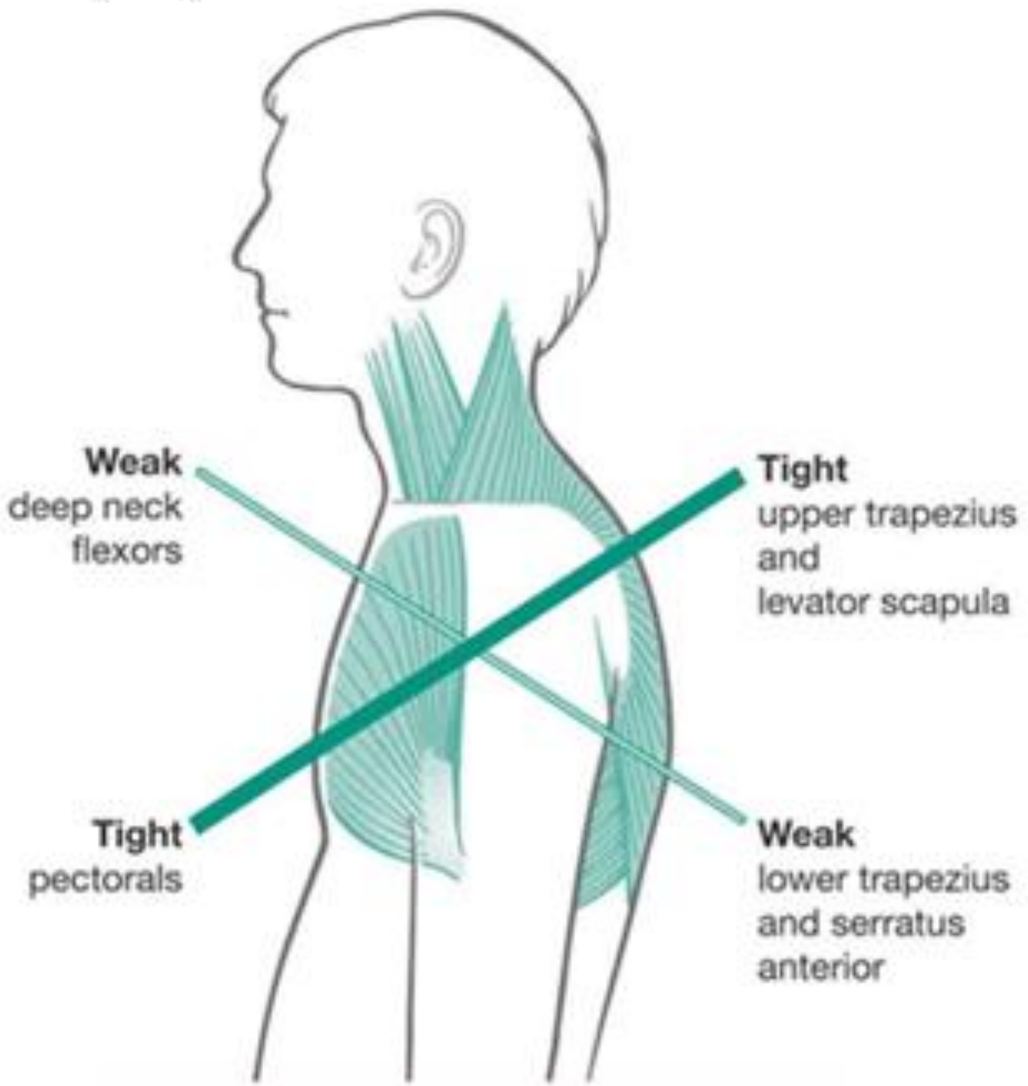
UPRIGHT POSTURE

a **problem** in one part
affects the other parts

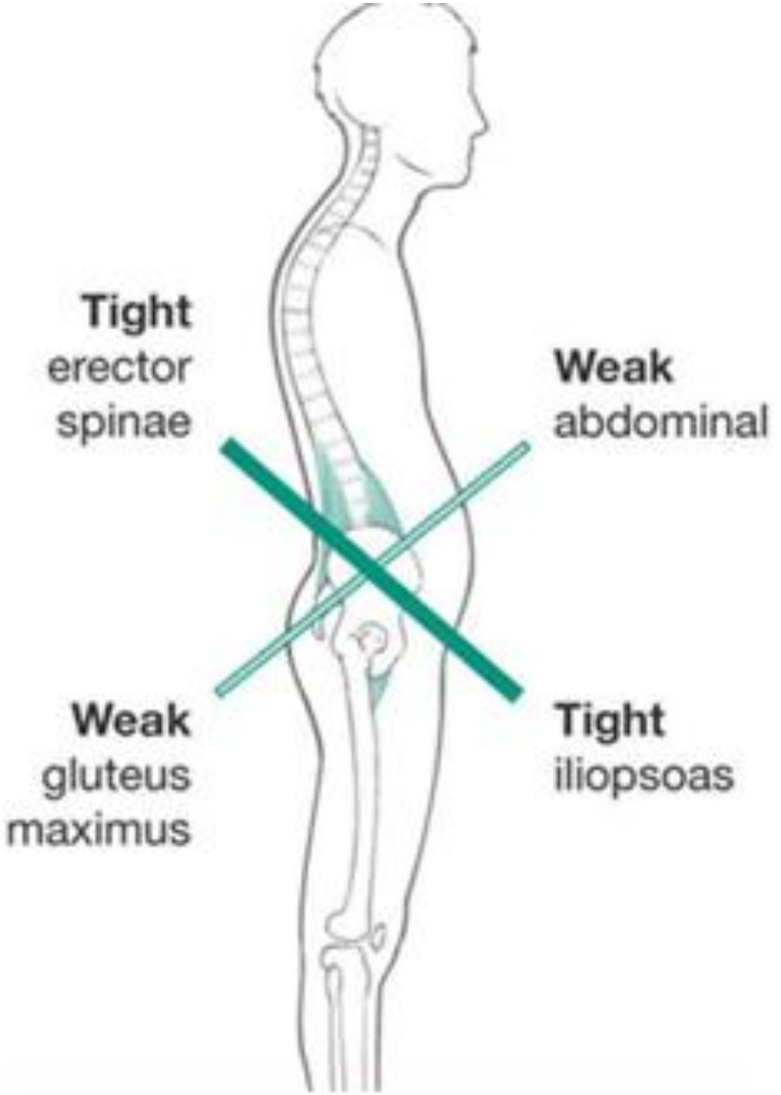
CROSS SYNDROME

symptoms of **tightness** and **weakness** on **one side** is **paired diagonally** across the **other side** of the region

neck and **lower back** area



Upper Crossed Syndrome

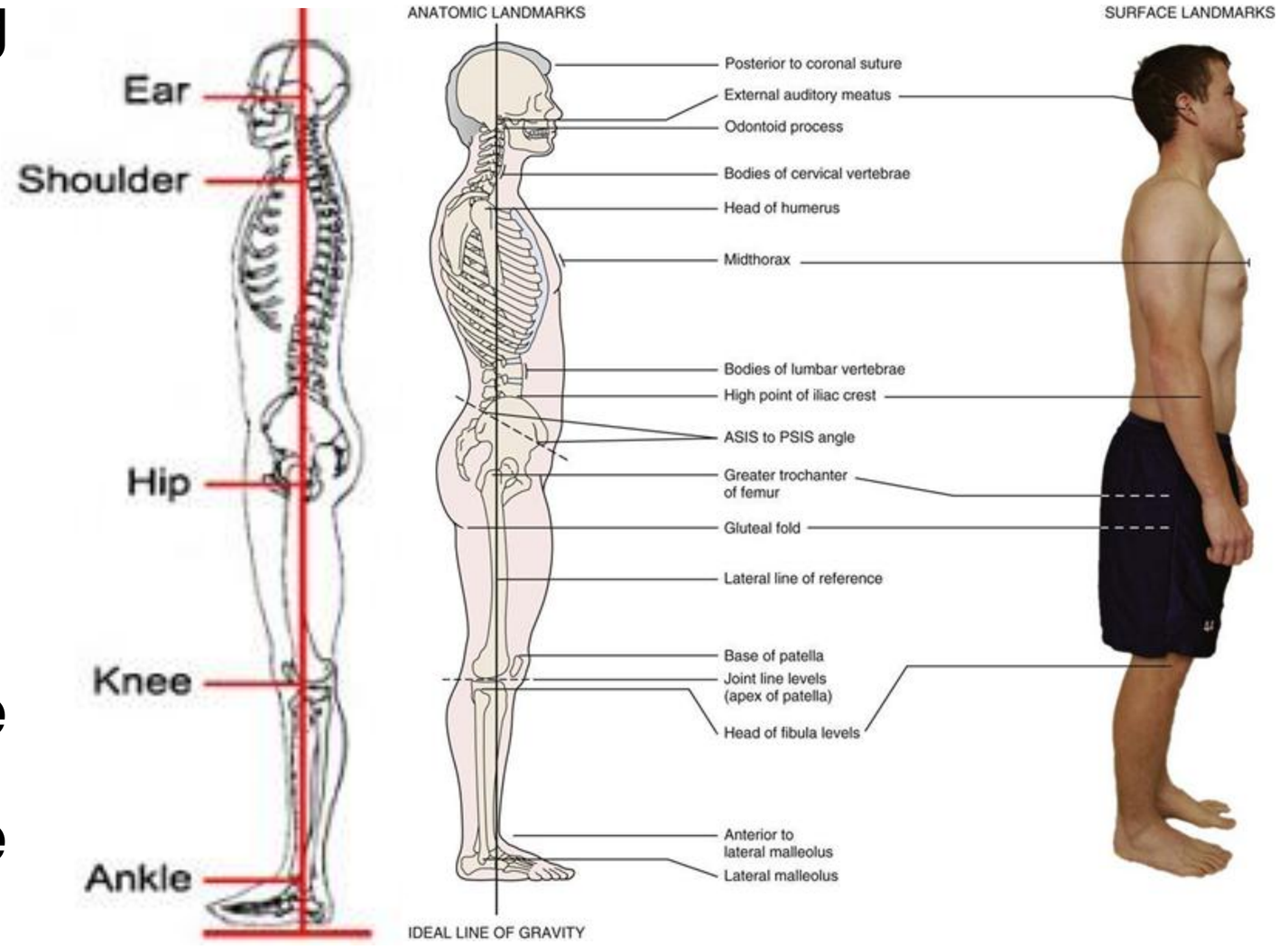


Lower Crossed Syndrome

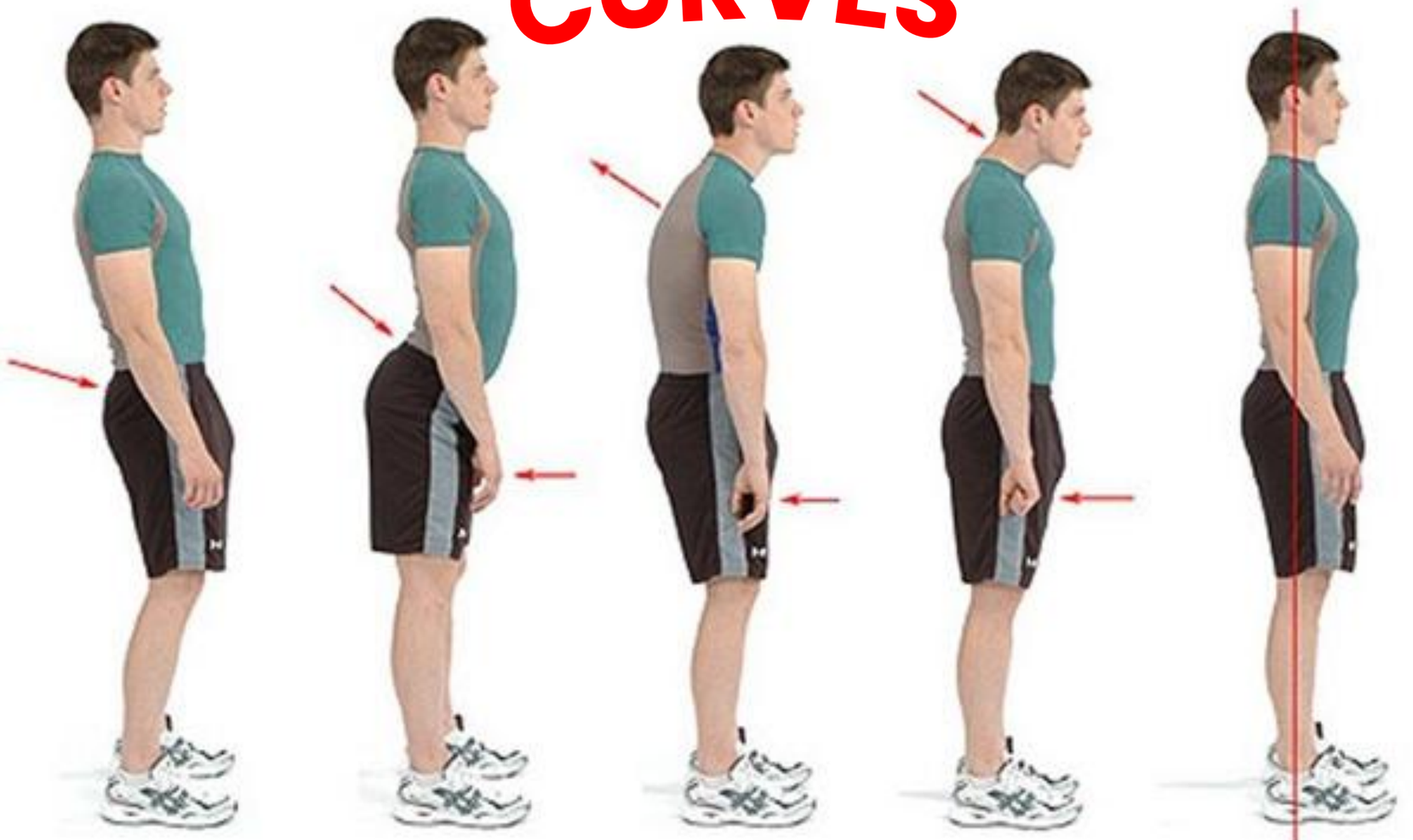
IDEAL POSTURE

a **line of gravity** passing through:

- Meatus
- Bodies of Cervical Vertebrae
- Acromion Process
- Middle of the Thorax
- Bodies of Lumbar Vertebrae
- Slightly Posterior to the Hip Joint
- Slightly Anterior to the Knee Joint
- Slightly Anterior to the Malleolus



CURVES



Sway Back

Lumbar Lordosis

Thoracic Kyphosis

Forward Head

Good Posture

What are the **implications** of having **GOOD**/**BAD** **Posture?**

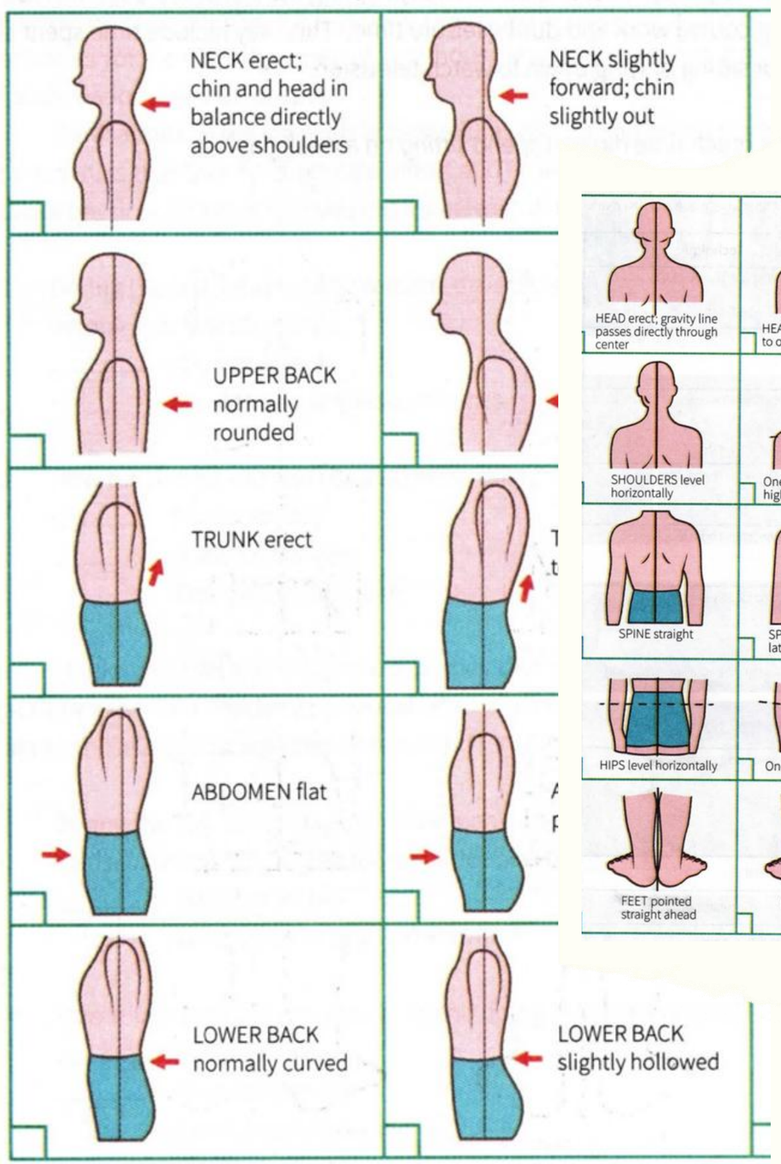
✓Minimal Stress-Load

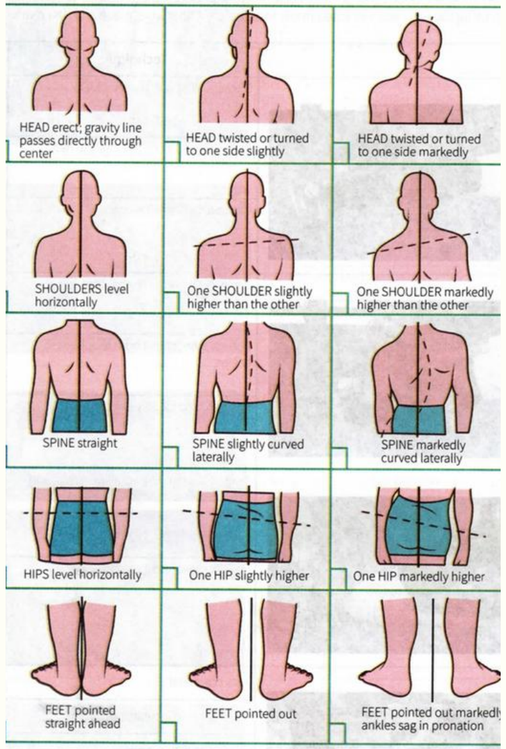
Increase in Muscle Activity = **BODY PAIN**

+++Accumulation of Stress

=

1. **HYPERMOBILE** (too flexible)
2. **HYPOMOBILE** (too stiff)





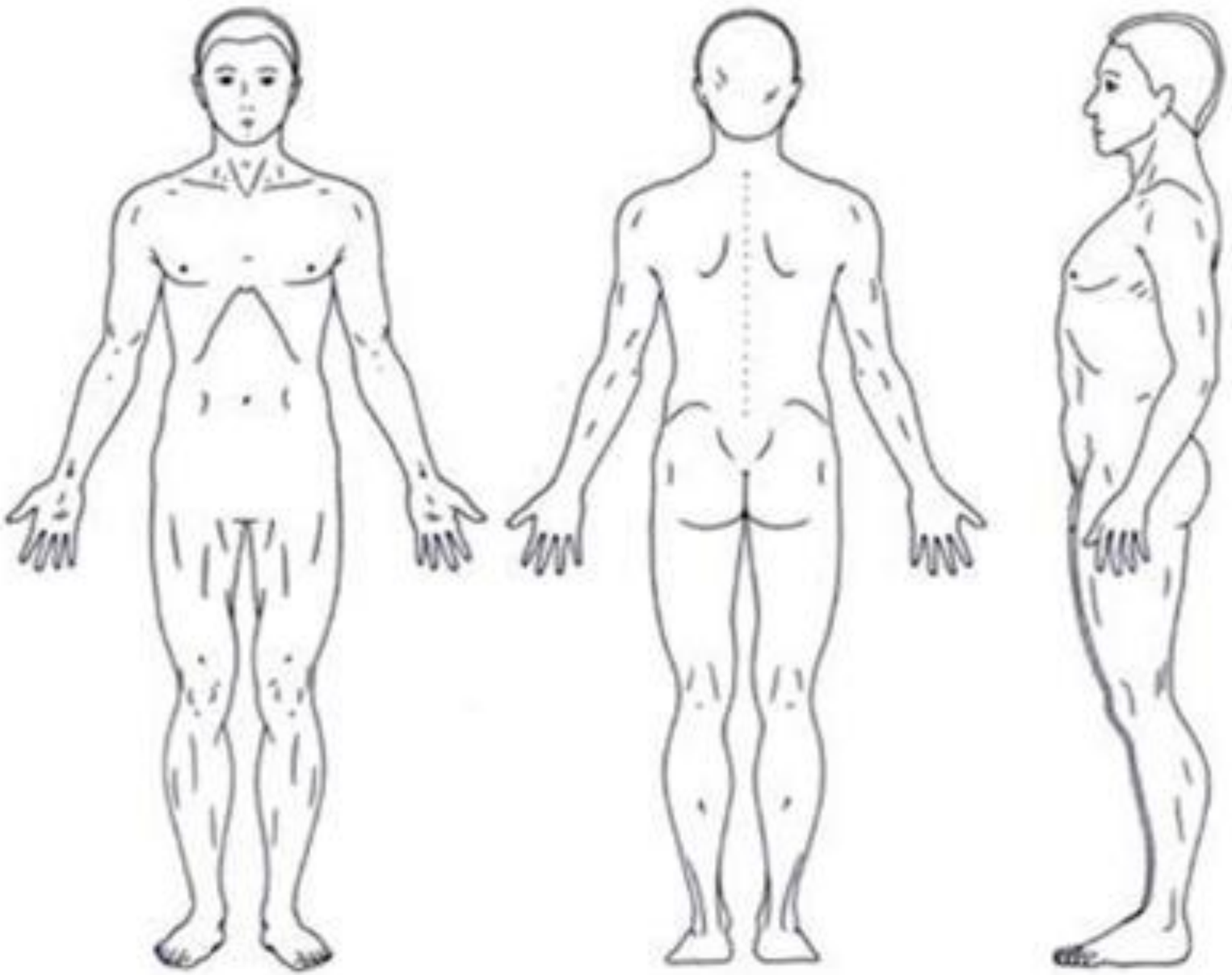
POSTURAL ASSESSMENT

UNDERSTANDING MOVEMENTS

terms and **references**

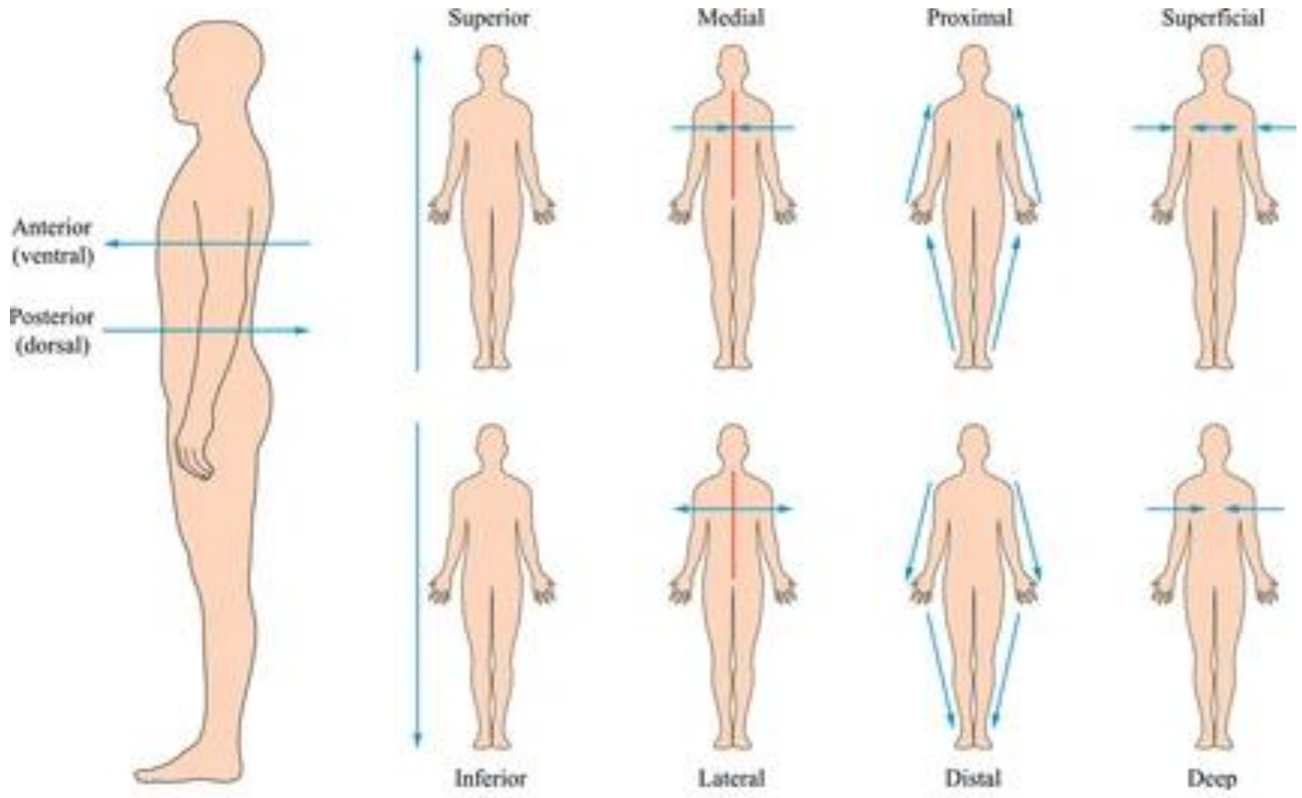
ANATOMICAL POSITION

a **standing position** with the **face, body, knees, and feet FACING FORWARD**, arms extended on the sides with the shoulders rotated so that the palms are also facing forward with the fingers extended.



DIRECTIONAL TERMS

- **Anterior** and **Posterior**
- **Superior** and **Inferior**
- **Lateral** and **Medial**
- **Proximal** and **Distal**
- **Superficial** and **Deep**
- **Supine** and **Prone**



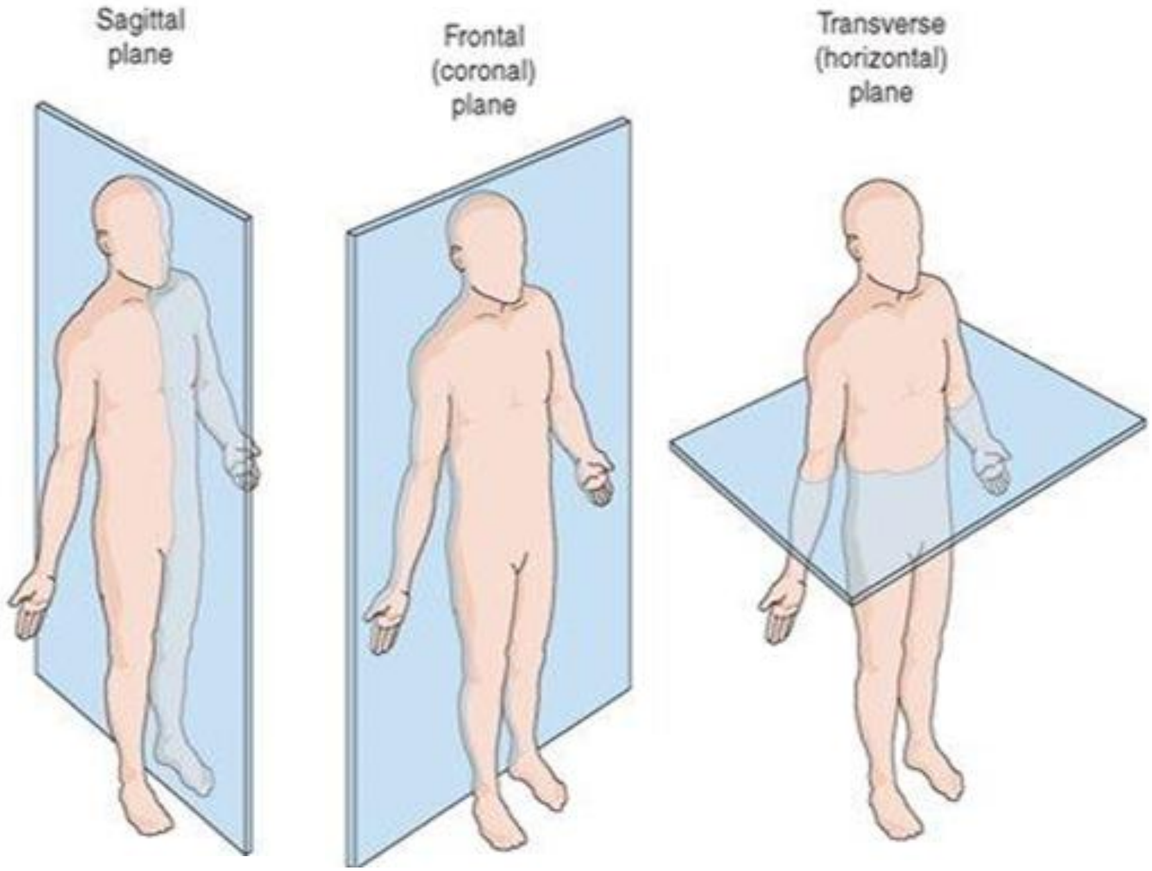
PLANES OF MOTION

imaginary planes | divide the body into different sections

S **SAGITTAL**
(right/left)

C **CORONAL**
(front/back)

T **TRANSVERSE**
(up/down)



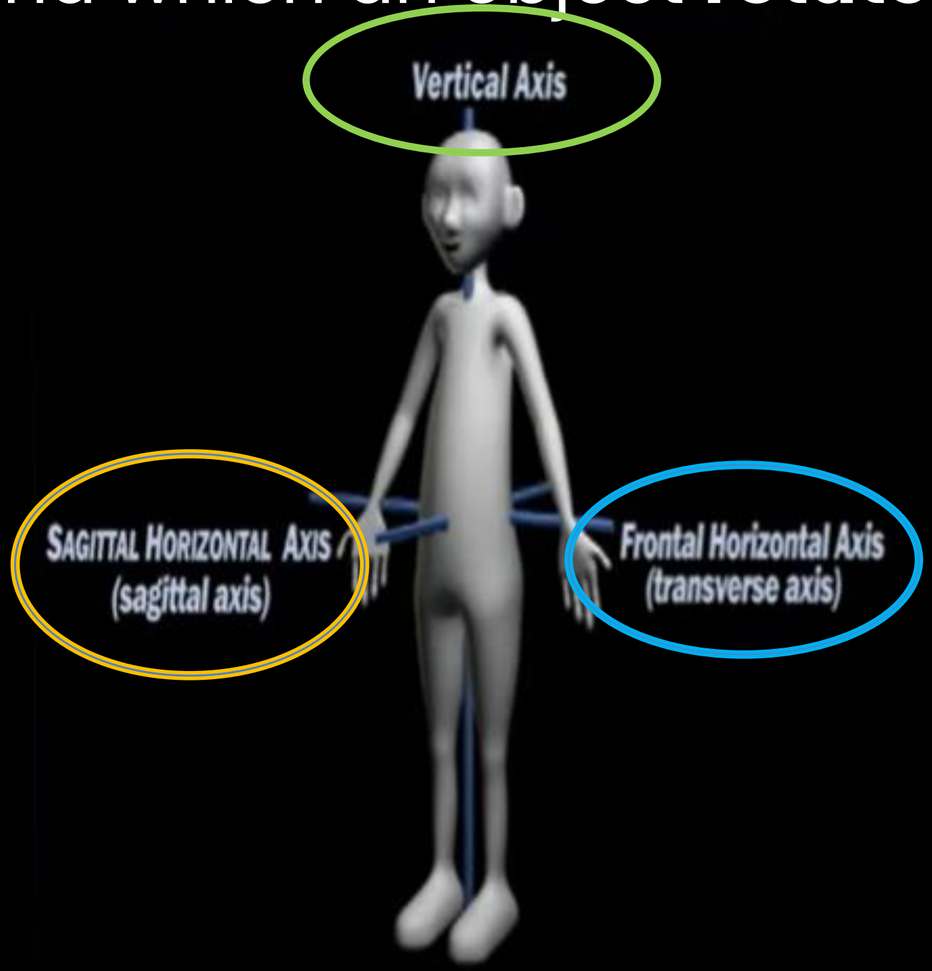
AXIS OF MOTION

an imaginary **straight line** around which an object **rotates**

SAGITTAL

VERTICAL

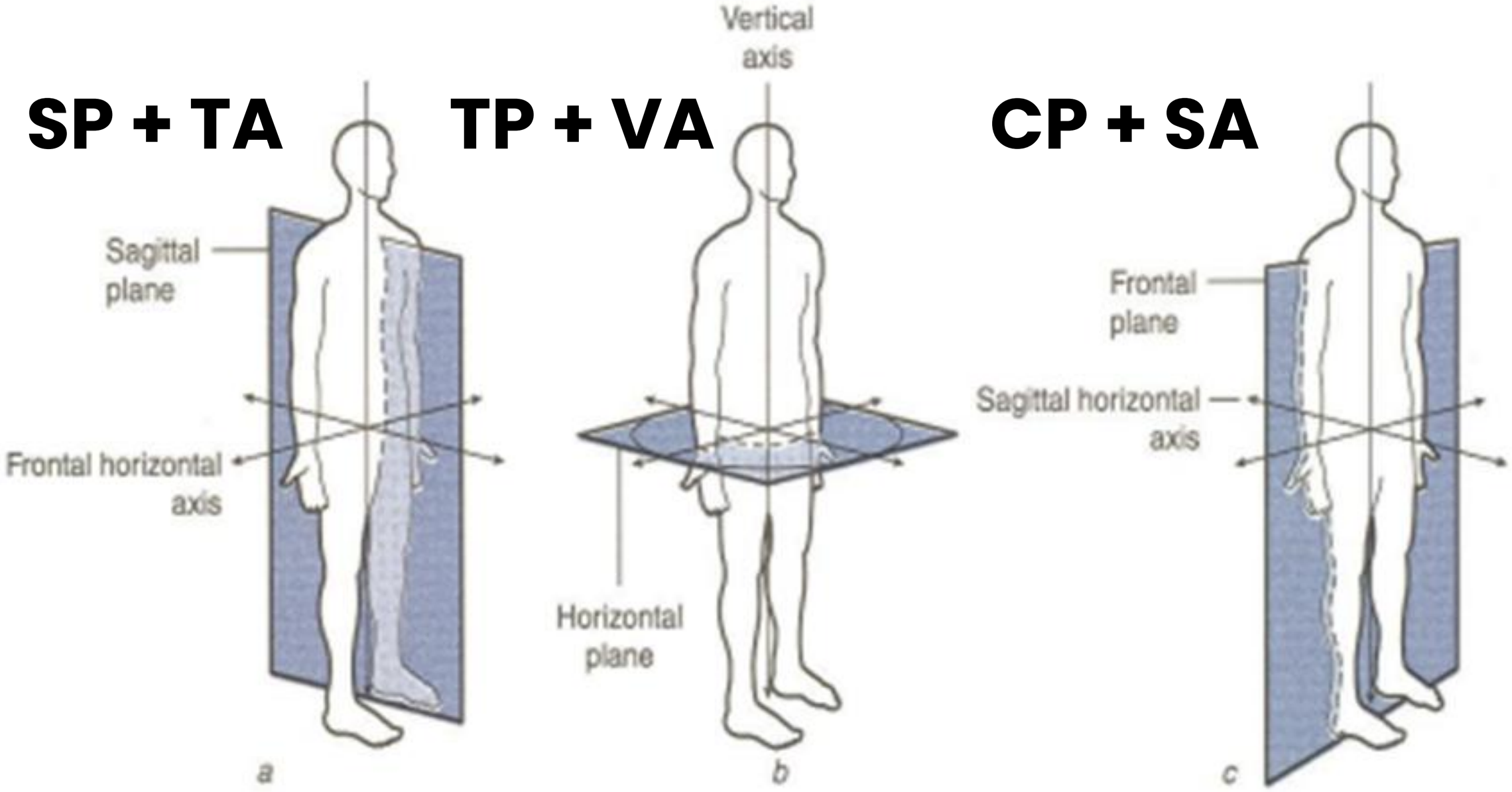
TRANSVERSE



SP + TA

TP + VA

CP + SA





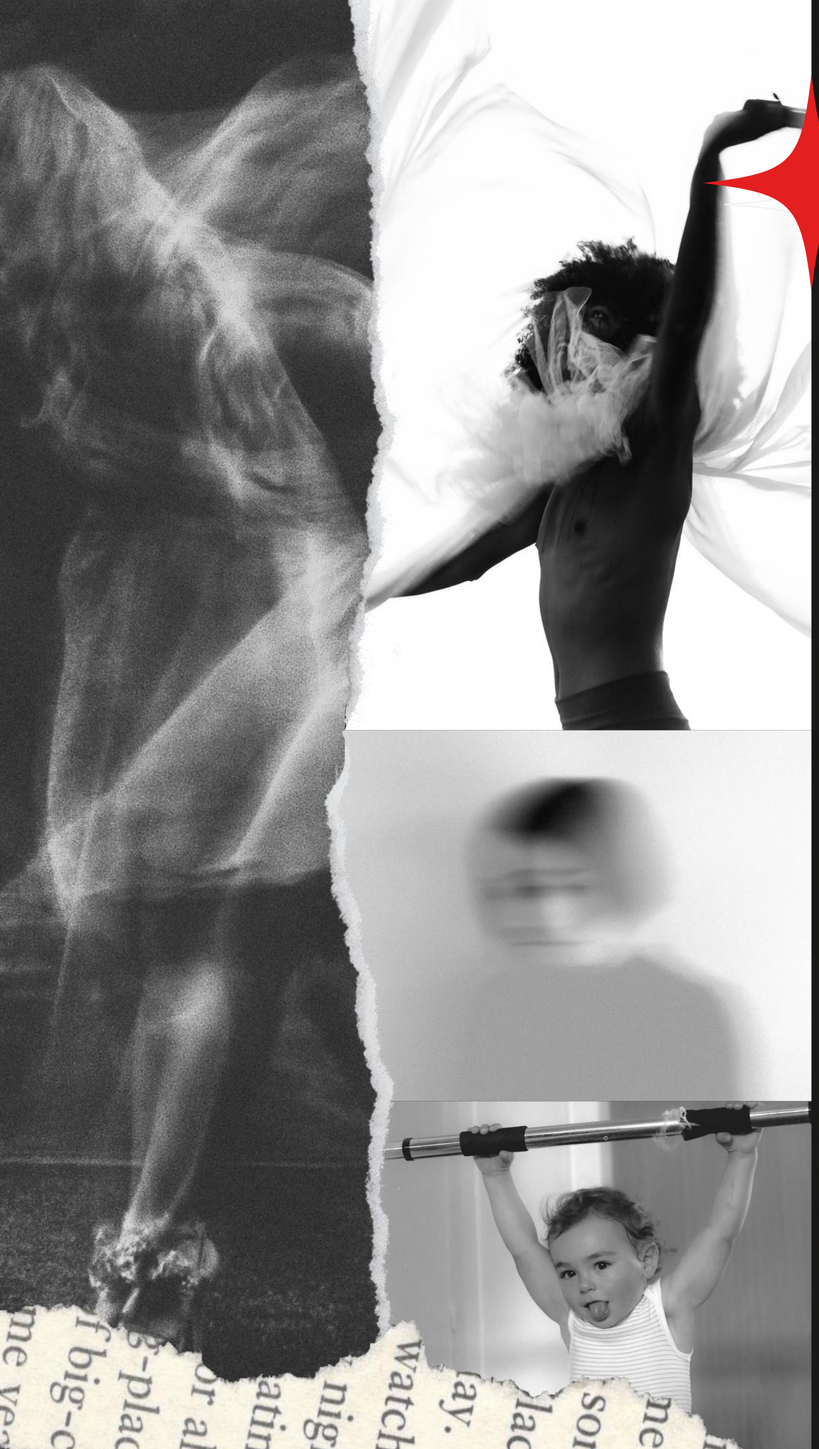




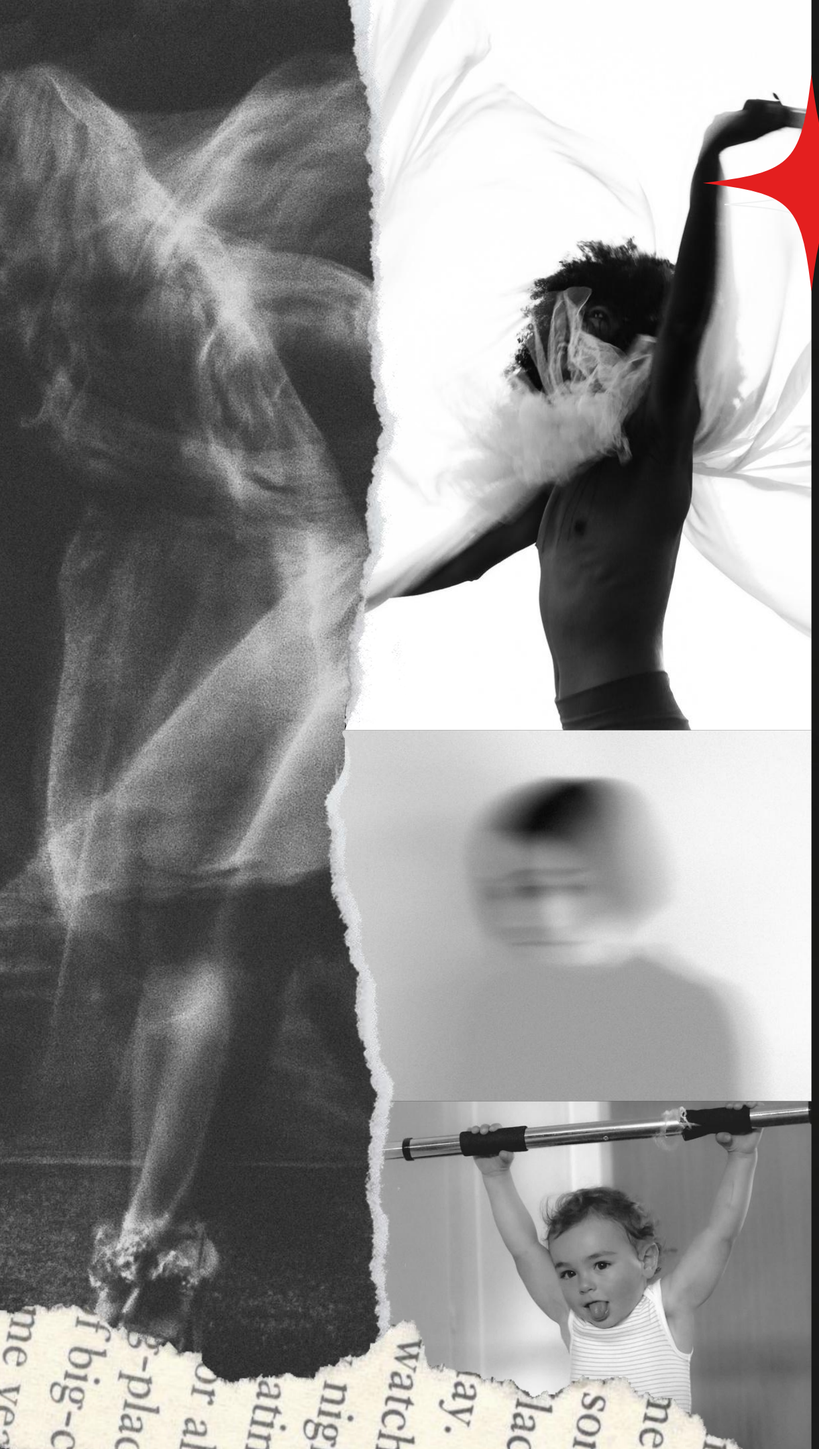


MOVEMENT TERMS

- **Flexion and Extension**
  Flexion is decreasing the angle between two segments while extension is increasing the angle.
- **Dorsiflexion and Plantar flexion**
  Flexion and extension on the ankle joint.
- **Inversion and Eversion**
  Also happens in the ankle joint.



- **Abduction and Adduction**
  Abduction is a movement away from the midline while adduction is a movement towards the midline.
- **Medial and Lateral Rotation**
  Rotating facing the midline and rotating facing the sides/lateral.
- **Pronation and Supination**
  Performed on the forearm causing the palms to face posteriorly and anteriorly.



- **Circumduction**
  A combination of flexion, extension, abduction, and adduction. It allows one to create big circular movements of the arms.
- **Protraction and Retraction**
  Means forward and backward and usually applied to the scapula or shoulder blades whereas protraction is moving the shoulders away from each other and retraction is moving closer together.









- **Depression and Elevation**
  A downward or upward movement of the scapula or mandible.

- **Opposition and Reposition**
  Movements of the thumb.

MOBILITY

the ability of the **JOINT** to move freely throughout its range of motion (ROM).

FUNCTIONS OF JOINTS

stability *mobility*

FUNCTIONS OF JOINTS

Ankle	Mobility
Knee	Stability
Hip	Mobility
Lumbar spine	Stability
Thoracic spine	Mobility
Scapula	Stability
Shoulder	Mobility

a **PROBLEM** at one point usually shows up as a **PAIN** in the **joint above or below**